

Chapter 1

Mission Bus Integration Lab (MBIL)

1.1 Project purpose

Mission Bus Integration Lab (MBIL) is a deterministic, configuration-driven software system that simulates a simplified aerospace mission environment built around a central data bus, redundant mission computers, subsystem messaging, defined failure modes, observability, and failover behaviour.

The goal is not to make a flashy simulator

The goal to demonstrate that I understand how **mission-critical systems behave, fail, recover, and get debugged.**

1.2 What MBIL should communicate

When someone sees this project, they should think:

- ★ This person understands reliable system design
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